

2013-2014 – Master 2 – Macro I

Lecture notes #13 : Solving Dynamic Rational Expectations Models

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Introduction

- ▶ Economic agents have to make a decision based on their forecast of some future variable.
- ▶ In the most general case, the entire probability distribution of that variable will affect their optimal choice.
- ▶ The rational expectations hypothesis states that people take into account the *actual* distribution of the relevant variable they want to forecast.
- ▶ If expectations are rational, we have to solve simultaneously for the equilibrium distribution and for the rest of the model
- ▶ We restrict to linear models in which the distribution of endogenous variables of interest only matters through its first moment, i.e. its mathematical expectation.

1. A simple example

Model

- ▶ Agricultural market where producers have to decide how much to produce prior to knowing the equilibrium price

$$y_t^d = a - bp_t + \varepsilon_{dt}$$

$$y_t^s = c + dp_t^a + \varepsilon_{st},$$

$$y_t^s = y_t^d.$$

- ▶ Here, p = price, ε_d = demand shock, ε_s = supply shock, y^s = supply, y^d = demand, p^a = anticipated price.

1. A simple example

Rational expectations

- ▶ The Rational Expectations Hypothesis assumes that

$$p_t^a = E(p_t \mid I_t),$$

where E = mathematical expectation operator and I_t = information set of the price-setters when they set y_t^s .

- ▶ Typically, $I_t =$
 {past values of all the variables, the model, how to solve it}
- ▶ If farmers set y_t^s at date $t - 1$, :

$$p_t^a = E_{t-1} p_t.$$

1. A simple example

Solving the model

- ▶ **General principle :**
 - ▶ take expectations in all the equations and then solve for the expected variables,
 - ▶ then substitute this in all the equations where expectations enter,
 - ▶ this allows to solve for the actual equilibrium variables.
- ▶ Let us try this here.

1. A simple example

Solving the model (continued)

$$\begin{aligned}y_t^d &= a - bp_t + \varepsilon_{dt} \\y_t^s &= c + dp_t^a + \varepsilon_{st}, \\y_t^s &= y_t^d.\end{aligned}$$

- ▶ The model can be collapsed to

$$a - bp_t + \varepsilon_{dt} = c + dE_{t-1}p_t + \varepsilon_{st}. \quad (1)$$

- ▶ Take expectations :

$$a - bE_{t-1}p_t + E_{t-1}\varepsilon_{dt} = c + dE_{t-1}p_t + E_{t-1}\varepsilon_{st}.$$

- ▶ Thus

$$E_{t-1}p_t = \frac{a - c + E_{t-1}(\varepsilon_{dt} - \varepsilon_{st})}{b + d}.$$

1. A simple example

Solving the model (continued)

- ▶ substitute this into (1) to solve for p_t :

$$p_t = \frac{1}{b}(\varepsilon_d - \varepsilon_s) - \frac{d}{b(b+d)}E_{t-1}(\varepsilon_d - \varepsilon_s) + \frac{a-c}{b+d}. \quad (2)$$

- ▶ This is the solution.

1. A simple example

The method of undetermined coefficients

- ▶ If we know the distribution of those shocks, we can use another method called the method of undetermined coefficients
- ▶ This is akin to the notion of Markov perfect equilibrium in game theory.
- ▶ Method :
 - ▶ Identify the relevant state space
 - ▶ Look for a solution as a (linear) function of this state vector
 - ▶ Given such a function, I can compute all the expectations I need
 - ▶ Substitute this function and the appropriate formula for expectations in all places where it is needed
 - ▶ Identify the coefficients on each state variable
 - ▶ This gives us equations in the coefficients that allow us to compute them

1. A simple example

The method of undetermined coefficients (continued)

- ▶ Assume $E_{t-1}\varepsilon_{dt} = E_{t-1}\varepsilon_{st} = 0$.
- ▶ The relevant state space is $(\varepsilon_d, \varepsilon_s)$.
- ▶ Look for a solution of the form

$$p_t = \alpha\varepsilon_{dt} + \beta\varepsilon_{st} + \gamma.$$

1. A simple example

The method of undetermined coefficients (continued)

- ▶ Under the assumed solution, we have

$$E_{t-1}p_t = \gamma.$$

- ▶ Substituting into (1) we get

$$a - b(\alpha\varepsilon_{dt} + \beta\varepsilon_{st} + \gamma) + \varepsilon_{dt} = c + d\gamma + \varepsilon_{st}.$$

- ▶ This must hold for all realizations of ε_d and ε_s and therefore we must have

$$\begin{aligned}a - b\gamma &= c + d\gamma \\ -\alpha b + 1 &= 0 \\ -\beta b &= 1.\end{aligned}$$

1. A simple example

The method of undetermined coefficients (continued)

- Therefore

$$\begin{aligned}\gamma &= \frac{a - c}{b + d} \\ \alpha &= 1/b \\ \beta &= -1/b.\end{aligned}$$

- Hence the solution

$$p_t = \frac{1}{b}(\varepsilon_d - \varepsilon_s) + \frac{a - c}{b + d},$$

- which is equivalent to (2) if $E_{t-1}\varepsilon_{dt} = E_{t-1}\varepsilon_{st} = 0$.

2. A dynamic model

Model

- Suppose now the supply curve is

$$y_t^s = c + dE_t p_{t+1} + \varepsilon_{st}$$

- Eliminating y between supply and demand :

$$p_t = \frac{a - c}{b} + \frac{-d}{b} E_t p_{t+1} + \frac{\varepsilon_{dt} - \varepsilon_{st}}{b}$$

- and assuming for simplicity that the constant terms are zero, we end up with

$$p_t = \rho E_t p_{t+1} + \eta_t, \tag{3}$$

where η_t is a shock.

2. A dynamic model

Solution by forward integration

- ▶ (3) implies $p_{t+i} = \rho E_t p_{t+i+1} + \eta_{t+i} \quad \forall i$
- ▶ Taking expectations

$$E_t p_{t+i} = \rho E_t p_{t+i+1} + E_t \eta_{t+i}$$

- ▶ Substitute forward (*forward integration*) :

$$p_t = \rho^k E_t p_{t+k} + \sum_{j=0}^{k-1} \rho^j E_t \eta_{t+j} \quad (4)$$

- ▶ Assume stationary shocks and impose $E_t p_{t+k}$ is bounded $\forall k$.
- ▶ (stochastic equivalent of selecting the non-explosive saddle-path as the unique solution.)
- ▶ Then if $|\rho| < 1$, then $\lim_{k \rightarrow \infty} \rho^k E_t p_{t+k} = 0$, hence

$$p_t = \sum_{j=0}^{\infty} \rho^j E_t \eta_{t+j}.$$

2. A dynamic model

Solution by forward integration (continued)

$$p_t = \sum_{j=0}^{\infty} \rho^j E_t \eta_{t+j}.$$

- ▶ This is the unique stationary solution to the rational expectations equation (3).
- ▶ Suppose $|\rho| > 1$, then the solution cannot be unique.
- ▶ For any solution p_t , $p_t + x_t$ is also a solution under some conditions on x
- ▶ x_t is a random variable (sunspot) satisfying

$$x_{t+1} = \frac{1}{\rho} x_t + u_{t+1}, \tag{5}$$

with u any shock such that $E_t u_{t+1} = 0$.

- ▶ Furthermore, we can construct at least one non explosive solution by picking $p_{t+1} = \frac{1}{\rho}(p_t - \eta_t)$.
- ▶ Hence in this case we have a continuum of equilibria.

2. A dynamic model

Undetermined coefficients

- ▶ We can apply the method of undetermined coefficients to this problem if we know the distribution of η .
- ▶ Assume η follows an AR1 :

$$\eta_t = \alpha\eta_{t-1} + \omega_t,$$

with ω_t iid and with zero mean.

- ▶ The only relevant state variables are η_{t-1} and ω_t so we look for a solution of the form

$$p_t = \lambda\eta_{t-1} + \beta\omega_t.$$

- ▶ Then

$$\begin{aligned} E_t p_{t+1} &= E_t(\lambda\eta_t + \beta\omega_{t+1}) \\ &= \lambda\eta_t \\ &= \lambda(\alpha\eta_{t-1} + \omega_t). \end{aligned}$$

2. A dynamic model

Undetermined coefficients (continued)

- We can then rewrite (3) as

$$\lambda\eta_{t-1} + \beta\omega_t = \rho\lambda(\alpha\eta_{t-1} + \omega_t) + \alpha\eta_{t-1} + \omega_t.$$

- Identifying, we get

$$\lambda = \rho\lambda\alpha + \alpha$$

$$\beta = \rho\lambda + 1$$

- Hence

$$\lambda = \frac{\alpha}{1 - \rho\alpha};$$

$$\beta = \frac{1}{1 - \rho\alpha}.$$

2. A dynamic model

Undetermined coefficients (continued)

- ▶ We can check that if $|\rho| < 1$, this is indeed correct :

$$\begin{aligned}\sum_{j=0}^{\infty} \rho^j E_t \eta_{t+j} &= \sum_{j=0}^{\infty} \rho^j \alpha^j \eta_t \\ &= \frac{1}{1 - \rho\alpha} (\alpha\eta_{t-1} + \omega_t) \\ &= \lambda\eta_{t-1} + \beta\omega_t\end{aligned}$$

- ▶ If $|\rho| > 1$, this is still a solution, and it is the unique "linear Markov" solution, i.e. the only one which is a linear combination of the relevant state space.
- ▶ But many other solutions can be constructed by adding a sunspot x satisfying (5).

3. A general model

Model

- ▶ General linear model (Blanchard-Kahn)

$$\begin{pmatrix} K_t \\ X_t \end{pmatrix} = A \begin{pmatrix} K_{t-1} \\ E_t X_{t+1} \end{pmatrix} + U_t, \quad (6)$$

- ▶ K_t is an $m \times 1$ vector of state (=predetermined) variables, meaning that K_{t-1} is fixed at the beginning of period t ,
- ▶ X_t is an $n \times 1$ vector of non-predetermined variables, i.e. expected future values of those variables, rather than lagged ones, affect current outcomes.
- ▶ A is an $(m+n) \times (m+n)$ matrix
- ▶ U an $(m+n) \times 1$ vector of shocks.

3. A general model

Model transformation

- Let

$$A = \begin{pmatrix} A_{00} & A_{01} \\ A_{10} & A_{11} \end{pmatrix}, \quad U_t = \begin{pmatrix} U_{0t} \\ U_{1t} \end{pmatrix},$$

- then

$$K_t = A_{00}K_{t-1} + A_{01}E_tX_{t+1} + U_{0t}$$

$$X_t = A_{10}K_{t-1} + A_{11}E_tX_{t+1} + U_{1t}.$$

- We can solve for E_tX_{t+1} and write the whole system with backward-looking dynamics :

$$\begin{aligned} \begin{pmatrix} K_t \\ E_tX_{t+1} \end{pmatrix} &= \begin{pmatrix} A_{00} - A_{01}A_{11}^{-1}A_{10} & A_{01}A_{11}^{-1} \\ -A_{11}^{-1}A_{10} & A_{11}^{-1} \end{pmatrix} \begin{pmatrix} K_{t-1} \\ X_t \end{pmatrix} \\ &+ \begin{pmatrix} U_{0t} - A_{01}A_{11}^{-1}U_{1t} \\ -A_{11}^{-1}U_{1t} \end{pmatrix} \\ &= M \begin{pmatrix} K_{t-1} \\ X_t \end{pmatrix} + V_t. \end{aligned} \tag{7}$$

3. A general model

The diagonal case

$$\begin{pmatrix} K_t \\ E_t X_{t+1} \end{pmatrix} = M \begin{pmatrix} K_{t-1} \\ X_t \end{pmatrix} + V_t. \quad (7)$$

- Let us start with the simple case where M is diagonal, say

$$M = \begin{pmatrix} D_0 & 0 \\ 0 & D_1 \end{pmatrix}, \text{ where } D_0 = \begin{pmatrix} d_1 & 0 & \dots & 0 \\ 0 & d_2 & \dots & \dots \\ \dots & \dots & \dots & 0 \\ 0 & \dots & 0 & d_m \end{pmatrix} \text{ and}$$

$$D_1 = \begin{pmatrix} d'_1 & 0 & \dots & 0 \\ 0 & d'_2 & \dots & \dots \\ \dots & \dots & \dots & 0 \\ 0 & \dots & 0 & d'_n \end{pmatrix} \text{ and } V_t = \begin{pmatrix} V_{0t} \\ V_{1t} \end{pmatrix}$$

3. A general model

The diagonal case (continued)

- ▶ Then, iterating (7),

$$K_{t+i} = D_0^{i+1} K_{t-1} + \sum_{j=0}^i D_0^{i-j} V_{0t+j}.$$

- ▶ For K_{t+i} to remain bounded, we need

$$|d_i| < 1, i = 1, \dots, m. \quad (8)$$

- ▶ Similarly

$$\begin{aligned} E_t X_{t+i} &= D_1^i X_t + \sum_{j=0}^{i-1} D_1^{i-1-j} E_t V_{1t+j} \\ &= D_1^i \left(X_t + \sum_{j=0}^{i-1} D_1^{-1-j} E_t V_{1t+j} \right). \end{aligned}$$

3. A general model

The diagonal case (continued)

- ▶ Consider first a k such that $|d'_k| < 1$. Let x_{kt} (resp. v_{kt}) be the k -th term in X_t (resp. V_{1t}).
- ▶ Then

$$\begin{aligned} E_t x_{t+i} &= (d'_k)^i x_t + \sum_{j=0}^{i-1} (d'_k)^{i-1-j} E_t v_{k,t+j} \\ &\iff (d_k'^{-1})^i E_t x_{t+i} - \sum_{j=0}^{i-1} (d_k'^{-1})^{1+j} E_t v_{k,t+j} = x_t. \end{aligned}$$

- ▶ This is similar to (4) and since $|d_k'^{-1}| > 1$ there is no unique initial value of x_t yielding a non-explosive path for subsequent expectations.

3. A general model

The diagonal case (continued)

- ▶ Second, assume conversely that

$$|d'_k| > 1 \text{ for } k = 1, \dots, n. \quad (9)$$

- ▶ Then all the terms in D_1^i go to infinity as $i \rightarrow \infty$.
- ▶ Thus it must be that $\lim_{i \rightarrow +\infty} X_t + \sum_{j=0}^{i-1} D_1^{-1-j} E_t V_{1t+j} = 0$.
Therefore,

$$X_t = - \sum_{j=0}^{\infty} (D_1^{-1})^{1+j} E_t V_{1t+j}.$$

3. A general model

The diagonal case (continued)

$$|d_i| < 1, i = 1, \dots, m. \quad (8)$$

$$|d'_k| > 1 \text{ for } k = 1, \dots, n. \quad (9)$$

- ▶ To summarize
 - ▶ To have a unique non-explosive solution, (8) and (9) must hold
 - ▶ If (8) is violated, all solutions are explosive
 - ▶ If (9) is violated, there is a continuum of non-explosive solutions.

3. A general model

The non diagonal case

- ▶ Assume now that the matrix M is not diagonal.
- ▶ The idea is to transform the model into a diagonal one.
- ▶ Generically, M is diagonalizable in the complex plane $\mathbb{C}$.
- ▶ Thus there exists P invertible such that

$$M = P^{-1}DP,$$

with D diagonal :

$$D = \begin{pmatrix} d_0 & 0 & \dots & \dots & \dots & \dots & 0 \\ 0 & \dots & \dots & \dots & \dots & \dots & \dots \\ \dots & \dots & d_m & 0 & \dots & \dots & \dots \\ \dots & \dots & 0 & d_{m+1} & 0 & \dots & \dots \\ \dots & \dots & \dots & 0 & \dots & \dots & \dots \\ \dots & \dots & \dots & \dots & \dots & \dots & 0 \\ 0 & \dots & \dots & \dots & \dots & 0 & d_{m+n} \end{pmatrix} = \begin{pmatrix} D_0 & 0 \\ 0 & D_1 \end{pmatrix}$$

3. A general model

The non diagonal case (continued)

- ▶ The diagonal coefficients are the eigenvalues of M .
- ▶ Eigenvalues are ranked by ascending module :
 $|d_0| \leq |d_1| \leq \dots \leq |d_m| \leq |d'_1| \leq |d'_2| \leq \dots \leq |d'_n|$.
- ▶ (7) rewrite

$$P \begin{pmatrix} K_t \\ E_t X_{t+1} \end{pmatrix} = D \left(P \begin{pmatrix} K_t \\ X_t \end{pmatrix} \right) + P V_t.$$

- ▶ Let $Z_t = P \begin{pmatrix} K_{t-1} \\ X_t \end{pmatrix}$;
- ▶ We are back to the diagonal case with this change of variable.

4. How general is the general model?

More lags for K

- ▶ Suppose that we have terms in K_{t-k} , $k > 1$.
- ▶ The solution is to add the variables $(K_{t-1}, \dots, K_{t-k+1})$ to the state vector.
- ▶ Suppose we have this equation $K_t = aK_{t-1} + bK_{t-k}$
- ▶ Let us assume to save on notation that K is a scalar.

- ▶ Then the vector $\begin{pmatrix} K_t \\ K_{t-1} \\ \dots \\ K_{t-k+1} \end{pmatrix}$ satisfies

$$\begin{pmatrix} K_t \\ K_{t-1} \\ \dots \\ K_{t-k+1} \end{pmatrix} = \begin{pmatrix} a & 0 & \dots & b \\ 1 & 0 & \dots & \dots \\ 0 & \dots & \dots & \dots \\ 0 & 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} K_{t-1} \\ K_{t-2} \\ \dots \\ K_{t-k} \end{pmatrix},$$

which only involves the first lag of this vector.

4. How general is the general model?

More lags for X

- ▶ Suppose we have terms in $E_t X_{t+k}$, $k > 1$.
- ▶ Let's add $E_t X_{t+1}, \dots, E_t X_{t+k-1}$ to the jump variables
- ▶ Suppose that we have

$$X_t = aE_t X_{t+1} + bE_t X_{t+k}.$$

- ▶ Then the vector $\psi_t = (X_t, E_t X_{t+1}, \dots, E_t X_{t+k-1})'$ satisfies

$$\begin{aligned}\psi_t &= \begin{pmatrix} X_t \\ E_t X_{t+1} \\ \dots \\ E_t X_{t+k-1} \end{pmatrix} = \begin{pmatrix} a & 0 & \dots & b \\ 1 & 0 & \dots & \dots \\ 0 & \dots & \dots & \dots \\ 0 & 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} E_t X_{t+1} \\ E_t X_{t+2} \\ \dots \\ E_t X_{t+k} \end{pmatrix} \\ &= \begin{pmatrix} a & 0 & \dots & b \\ 1 & 0 & \dots & \dots \\ 0 & \dots & \dots & \dots \\ 0 & 0 & 1 & 0 \end{pmatrix} E_t \psi_{t+1}.\end{aligned}$$